

A Projection-First Reframing of String Theory: Conditional Encodings, Worldsheet Gates, and the q79 Boundary

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Version 2

Abstract

String theory supplies a highly constrained language for extended excitations, quantum gravity, gauge sectors, duality, and compactification. A projection-first viewpoint can organize these structures as possible encodings of a richer source, but projection alone does not force strings, branes, a worldsheet, a critical dimension, modular invariance, or a graviton. This paper separates the internal results of perturbative string theory from their proposed interpretation in Modal Triplet Theory. A valid MTT-to-string bridge must construct target fields, a worldsheet action and measure, Weyl and modular consistency, anomaly cancellation, factorization, and the relevant observable dictionary from the same selected source. The q79 Fu–Yau branch is the present compatible candidate. Its twelve-row heterotic worldsheet contract has five available rows, two partial rows, and five open rows. The resulting claim is a structured string-encoding program, not a derivation of string theory from projection or a completed ultraviolet theory.

Version 2 Revision Note

Supersedes	Version 1.0, <i>A Projection-First Reframing of String Theory</i> .
Reason	The original paper presented strings, closed-string gravity, D-branes, holography, and the rejection of most vacua as consequences of finite admissibility. Those conclusions require independent worldsheet, boundary, anomaly, and observable structures.
Resolution	String structures are now conditional realization classes. The paper distinguishes standard string-theoretic results from an MTT interpretation and states the current q79 worldsheet contract row by row.
Retained result	Projection-first language remains useful for asking whether several lower descriptions descend from one selected source and preserve the same physical data.
Remaining boundary	The physical q79 bundle, exact $(0, 2)$ theory, modular/GSO completion, q79 string-field vertices, infrared completion, and nonperturbative definition remain open.

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1 What is being reframed

String theory is not merely the assertion that fundamental objects are one-dimensional. Its predictive structure comes from a linked package:

- a two-dimensional worldsheet field theory;
- target-space fields and boundary conditions;
- gauge fixing, ghosts, and a quantum measure;
- Weyl and modular consistency;
- GSO projection, factorization, and anomaly cancellation;
- a spectrum and observable prescription; and
- compactification and effective-field-theory limits.

Extended carriers are one part of this package. A loop drawn in a source space is not yet a string worldsheet, and a finite projector is not yet a worldsheet path integral.

The projection-first proposal is interpretive. It asks whether string variables can be a faithful lower description of selected upper data:

$$\Pi_{\text{ws}} : \mathcal{K}_{\text{MTT}} \longrightarrow \mathcal{C}_{\text{ws}}.$$

The codomain must contain more than embeddings $X^\mu(\sigma)$. It must contain all data needed to define the quantum worldsheet theory and its target-space interpretation.

2 What standard string theory establishes

2.1 Closed strings and gravity

Within perturbative closed-string theory, quantization of the worldsheet produces a massless spin-two state. Consistency of its interactions leads to general covariance and, at low energy, an Einstein-type target-space action with additional fields and higher-derivative corrections [1, 2]. This is a strong internal result of the specified string construction.

It is not a consequence of a generic many-to-one map. A projection can have no spin-two sector at all. An MTT bridge must therefore show that its selected source maps to the relevant closed-string Hilbert space, BRST cohomology, three-point couplings, and low-energy normalization.

2.2 Open strings and D-branes

D-branes are boundary conditions for open strings and carry tension, Ramond–Ramond charge, worldvolume fields, and consistency conditions. Gauge fields on brane stacks arise from open-string sectors and Chan–Paton data [3]. Calling a boundary an “admissibility boundary” can be a useful analogy, but it does not supply these charges, spectra, or amplitudes.

To realize a D-brane from MTT one must construct:

$$\begin{aligned} &\text{boundary condition} + \text{open-string state space} + \text{charge} \\ &\quad + \text{worldvolume action} + \text{anomaly conditions.} \end{aligned}$$

Without this contract the correspondence is interpretive only.

2.3 Criticality and modular consistency

Critical dimensions in perturbative string theories arise from cancellation of the worldsheet conformal anomaly for a declared matter and ghost system. They are not dimensional consequences of a circle, a rank flag, or a projection. Likewise, modular invariance is a property of a torus partition function and its state sum. A finite arithmetic symmetry may support a candidate construction, but it does not replace analytic characters, GSO projection, or factorization.

3 What projection-first language can claim

Projection-first language is strongest when used as a typing discipline. Suppose an upper object $\mathcal{U}$ carries connections, actions, states, and observables. A string encoding is physically meaningful only if maps

$$\Pi_{\text{target}}, \quad \Pi_{\text{ws}}, \quad \Pi_{\text{obs}}$$

preserve the structures required on their common domain. For example, a background equation obtained from a target-space action and a beta function obtained from the worldsheet must agree after conventions and field redefinitions are fixed. Similar-looking equations are not enough.

This supports a conditional statement:

If one selected upper construction admits a complete, anomaly-free worldsheet realization and a controlled target-space descent, then string theory can be a lower encoding of that construction on the common domain.

It does not support the converse claim that finite admissibility forces strings or that every consistent string vacuum is physically selected.

4 Why extended carriers are not automatic

A nontrivial loop or holonomy requires at least a suitable one-dimensional cycle. This explains why circles and line phases recur in Fourier analysis, gauge theory, and string compactification. It does not select a dynamical string. A point, graph, matrix algebra, particle worldline, higher-dimensional brane, or noncommutative algebra can also carry finite constraints.

Likewise, indecomposability of a constraint system does not imply an extended carrier. The MTT saturation paper gives finite countermodels: complete and locally rigid constraint systems can exist on point-valued carriers. A string-like realization becomes preferred only after additional requirements such as continuous loop support, reparametrization symmetry, worldsheet locality, or anomaly cancellation are imposed.

The proper hierarchy is therefore:

$$\text{projection} \not\Rightarrow \text{extended carrier} \not\Rightarrow \text{worldsheet QFT} \not\Rightarrow \text{consistent string vacuum}.$$

Each arrow requires a construction.

5 Duality, holography, and the landscape

5.1 Duality

A duality requires two theories, an invertible comparison on the declared domain, and preservation of observables or amplitudes. The shared-circle Fourier structure can motivate a T -duality candidate, but exact string-theoretic T -duality also acts on momentum, winding, background fields, and quantum states. S -duality requires an additional strong–weak coupling map. Neither follows from the word “projection.”

5.2 Holography

Holographic duality is not simply the survival of information on a boundary. It needs two fully defined theories and a quantitative dictionary equating partition functions, states, or correlators. Projection-first language can classify bulk and boundary variables, but it does not prove AdS/CFT or a general holographic principle.

5.3 Vacua and selection

The string landscape is a set of candidate compactifications subject to increasingly strong consistency conditions. MTT may add a source-selection criterion and correlate data ordinarily chosen independently. Yet mathematical consistency, local stability, cosmological realization, and empirical adequacy are different filters. No current MTT theorem eliminates all but one physical string vacuum.

6 The selected q79 candidate

The strongest current string-compatible MTT branch is the q79 Fu–Yau candidate. Its role should be stated carefully:

- the q79 arithmetic and finite carrier data are exact at their declared tier;
- Fu–Yau and related torus-bundle solutions provide established Hull–Strominger geometry [4, 5];

- a compatible finite Green–Schwarz/Bianchi representative and curvature-level visible cancellation are available;
- the physical nonpullback visible and hidden bundles and their continuum HYM realization are not yet selected; and
- no exact q79 $(0, 2)$ SCFT or nonperturbative string definition has been constructed.

The auxiliary literal product $S^1 \times \text{Lens} \times \text{Nil}$ is not the q79 compactification. Circle–Lens–Nil language is retained as a rank/operator filtration; the physical compactification candidate is the q79 Fu–Yau branch.

6.1 The twelve-row worldsheet contract

Row	Required object	Current status
W1	Time-oriented q79 target branch	Available
W2	Fu–Yau Mukai charge and Green–Schwarz/Bianchi sector	Available
W3	Visible curvature-level Green–Schwarz cancellation	Available at curvature tier
W4	Critical heterotic central-charge balance	Available universally
W5	q79 low-energy GR and quantum-EFT limit	Available at declared parity tier
W6	Global Deligne gerbe and Freed–Witten condition on the full visible cycle set	Open; finite representative only
W7	All-orders-in- α' q79 target background	Open; current background is first order
W8	Exact q79 heterotic $(0, 2)$ SCFT or all beta functions	Partial; base GLSM, local anomaly, and topological candidate exist, but the physical bundle and IR SCFT do not
W9	q79 modular-invariant GSO partition function and factorization	Partial; finite torsion module and seven-seed induction exist, but analytic characters and GSO remain open
W10	q79-specific string-field vertices and BV master action	Open; standard framework exists but is not instantiated on q79
W11	Tadpole, vacuum-shift, infrared, and soft-state completion	Open
W12	All-genus convergence or a nonperturbative definition	Open

The authoritative count is therefore five available, two partial, and five open. Standard heterotic string-field BV machinery exists [6]; importing that framework is not the same as constructing its q79 vertices.

7 Claim ledger

Claim	Status	Boundary
Strings are allowed MTT encodings	Supported	Requires a declared bridge and domain.
Projection forces strings	False in general	Counterexamples include finite point-valued carriers.
Closed strings contain a graviton	Standard string result	Assumes the quantized closed-string construction.
MTT derives the closed-string sector	Open	Needs state-space, BRST, and coupling intertwiners.
D-branes are admissibility boundaries	Interpretive analogy	Charges, boundary CFT, and worldvolume dynamics are additional.
q79 selects a compatible string route	Supported	Fu–Yau heterotic candidate at partial contract tier.
q79 defines a complete worldsheet theory	Not established	Current contract is 5 available, 2 partial, 5 open.
MTT uniquely selects a physical vacuum	Open	Needs complete source, dynamics, and observable tests.

8 Discussion

The projection-first interpretation remains valuable after these restrictions. It changes the research question from “are strings literally fundamental?” to “which upper data can a string description encode, and with what loss?” That question can be tested map by map.

The strongest route is not to redescribe familiar string motifs in new language. It is to complete the q79 contract and then compare two independently constructed descendants of the same source: the four-dimensional effective theory and the worldsheet theory. Agreement of actions, anomalies, states, and observables would be substantive evidence for a common encoding. Failure of a row would identify exactly where the analogy stops.

9 Conclusion

String theory is a possible and unusually rich projection-compatible encoding; it is not a consequence of projection alone. The extended carrier, worldsheet quantum theory, criticality, branes, dualities, and observables must each be supplied.

MTT has selected exact finite q79 data and a compatible Fu–Yau route, but the physical string construction remains incomplete. The next decisive steps are the visible-hidden HYM bundles, the exact $(0, 2)$ infrared theory, modular/GSO completion, and q79-specific string-field vertices. Until those are built, the honest result is a sharply organized bridge program rather than a derivation of string theory or a finished ultraviolet completion.

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